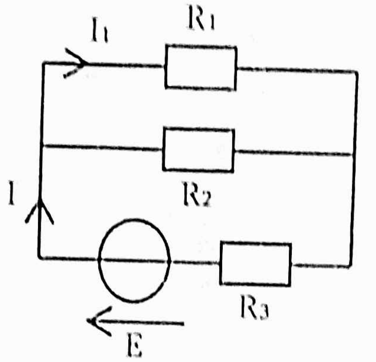
Université Sidi Mohammed Ben Abdallah
Faculté des Sciences et Techniques Fès
Département de Génie Electrique

Février 2024

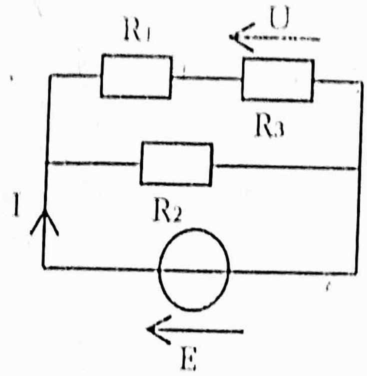
Série 2 S2 MIP

Exercice : 1

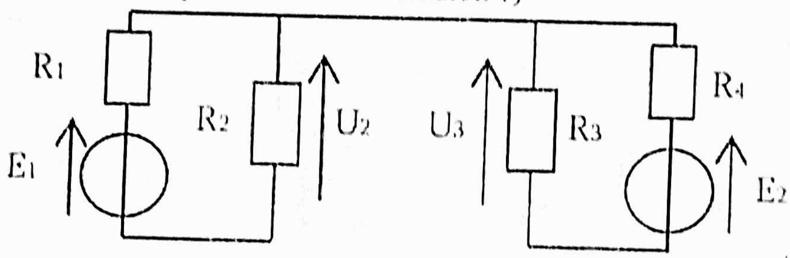
a) Déterminer I et I_1



b) Déterminer I et U



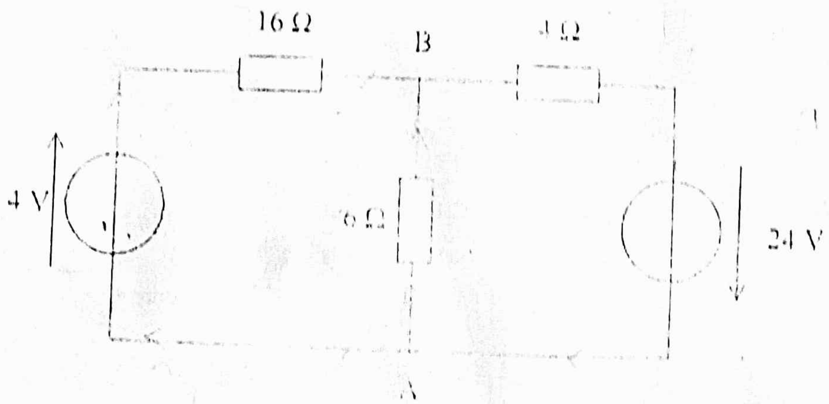
c) Déterminer U2 et U3 en fonction de E1, E2, R1, R2, R3 et R4
(Attention, peut-on utiliser le pont diviseur de tension ?)



Exercice : 2

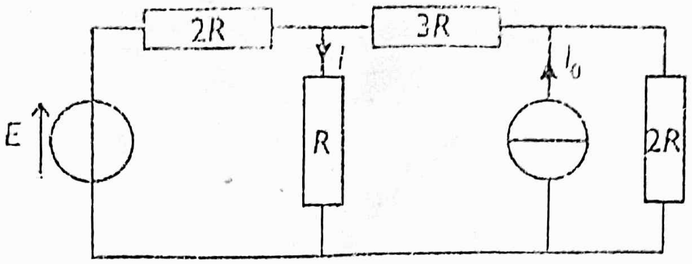
Calculer l'intensité du courant dans la branche AB en appliquant (circuit ci-dessous):

- les lois de Kirchhoff
- le théorème de Millman
- le théorème de superposition.

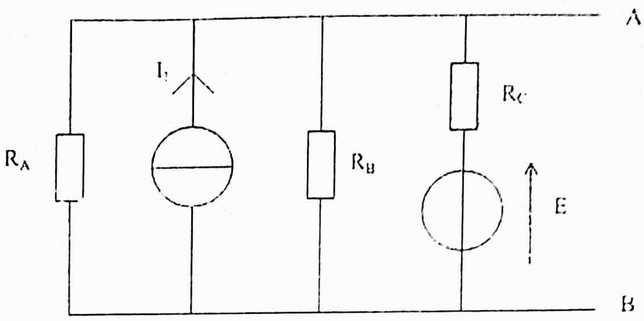


Exercice : 3

1. Remplacer le générateur de Norton (dans le circuit) par un générateur de Thévenin.
2. Déterminer l'intensité I du courant circulant dans la résistance R en utilisant :
   a- Les lois de Kirchhoff.
   b- Les Théorèmes de Thévenin et de Norton.
   c- La transformation Thévenin-Norton.
   d- La loi des nœuds en termes de potentiels (ou théorème de Millman).

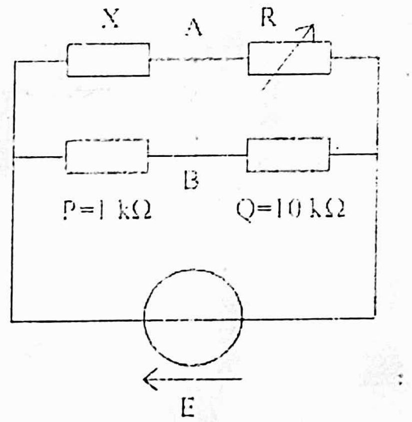


Exercice 4: Déterminer les modèles de Thévenin et de Norton du circuit suivant :

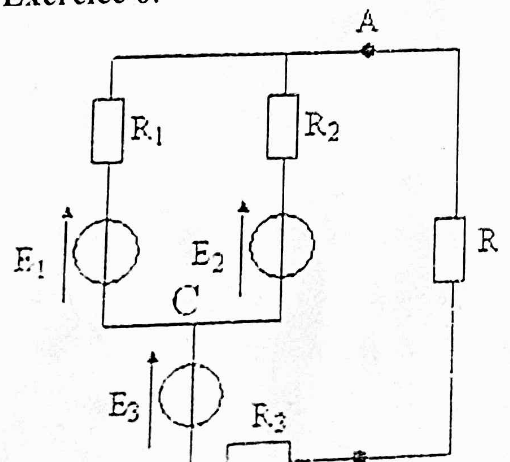


A.N. E = 12 V, I1 = 3 mA, RA = 1,5 kΩ, RB = 1 kΩ et RC = 3 kΩ.

Exercice 5:
Déterminer le modèle de Thévenin du dipôle AB.(On peut utiliser le diviseur de tension)
A quelle condition sur R a-t-on $U_{AB} = 0$ V ?
A.N :U_{AB} s'annule pour R = 8,75 kΩ. En déduire X la valeur de la résistance inconnue.



Exercice 6:

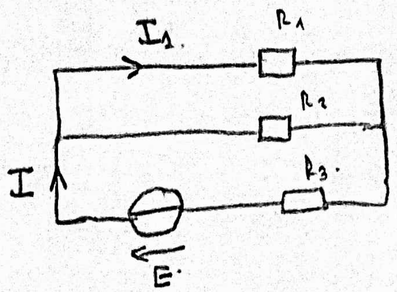


On considère le réseau représenté par le schéma ci-contre,
Calculer l'intensité du courant dans la résistance R
1°) En utilisant le théorème de Thévenin,
On donne : E_1= 3V ; R_1=R_2=R_3= 2Ω
E_2= 1V ; E_3= 2V ; R= 5Ω
2°) En utilisant le théorème de Norton
3°) Utiliser la transformation Thévenin-Norton
4°) En utilisant le théorème de Millman
5°) En utilisant les lois de Kirchhoff (facultative).

Série 2 Circuits

Exercice 1.

a) $I = \frac{E}{R_{eq}}$



$$\boxed{I = \frac{E}{\frac{R_1 R_2}{R_1+R_2} + R_3}}$$

$$\frac{1}{R_{eq_1}} = \frac{1}{R_1} + \frac{1}{R_2}$$

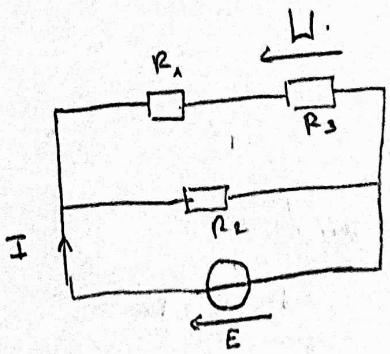
$$\frac{1}{R_{eq_1}} = \frac{R_2 + R_1}{R_1 R_2}$$

$$\boxed{R_{eq_1} = \frac{R_1 R_2}{R_1 + R_2}}$$

$$R_{eq} = R_{eq_1} + R_3$$

$$I_1 = \frac{R_2}{R_2 + R_1} I$$

b) $I = \frac{E}{R_{eq}}$



$$= \frac{E}{\frac{R_2(R_1+R_3)}{R_2+R_1+R_3}}$$

$$\boxed{I = \frac{R_2 + R_1 + R_3}{R_2(R_1+R_3)} E}$$

$$R_{eq_1} = R_1 + R_3$$

$$\frac{1}{R_{eq}} = \frac{1}{R_2} + \frac{1}{R_{eq_1}}$$

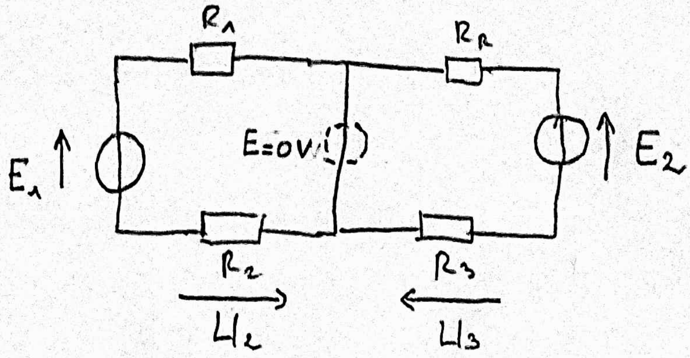
$$R_{eq} = \frac{R_2 \times R_{e1}}{R_2 + R_{eq_1}} = \frac{R_2(R_1+R_3)}{R_2 + R_1 + R_3}$$

$$\boxed{U = \frac{R_3}{R_1+R_3} E}$$

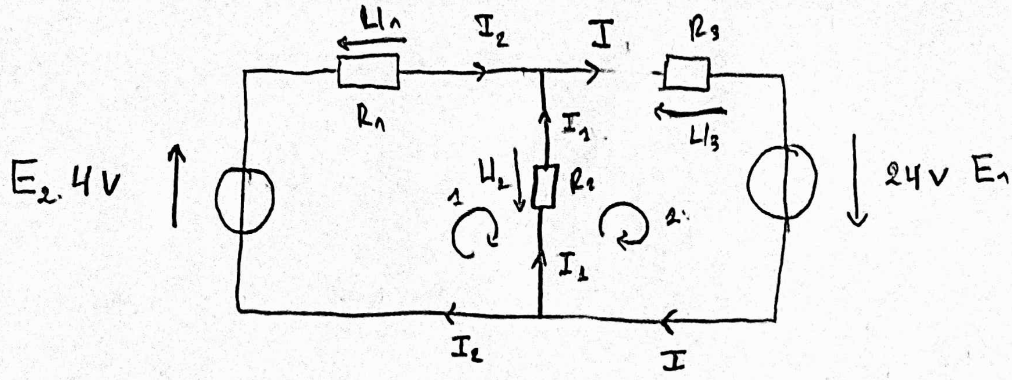
c)

$$U_2 = \frac{R_2}{R_1+R_2} \cdot E_1$$

$$U_3 = \frac{R_3}{R_3+R_4} \cdot E_2$$



Exercice 2.



1) • Calculer I_1

• Lois de Kirchhoff

1) Loi des noeuds

$$I = I_1 + I_2 \qquad (1)$$

• Loi des mailles

maille 1. => $E_2 - U_1 + U_2 = 0. \qquad (2)$

maille 2 => $E_1 - U_2 - U_3 = 0. \qquad (3)$

$$\begin{array}{l} L_1 \\ L_2 \\ L_3 \end{array} \left\{ \begin{array}{l} I_1 = I - I_2 \\ E_2 - I_2R_1 + I_1R_2 = 0 \\ E_1 - I_1R_2 - IR_3 = 0 \end{array} \right.$$

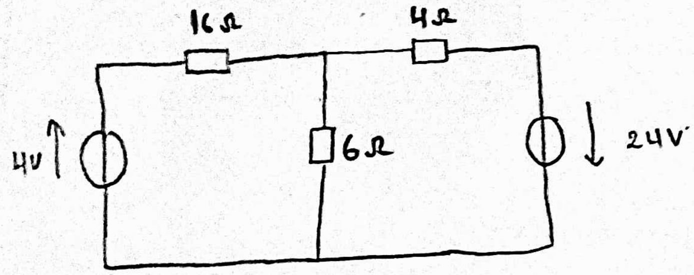
$$\begin{array}{l} L_1 \\ R_3 \times L_2 \\ -R_1 \times L_3 \end{array} \left\{ \begin{array}{l} I_1 = I - I_2 \\ R_3E_2 - I_2R_1R_3 + I_1R_2R_3 = 0 \\ -R_1E_1 + I_1R_2R_1 + IR_3R_1 = 0 \end{array} \right.$$

$$\begin{array}{l} L_1 \\ L_2+L_3 \end{array} \left\{ \begin{array}{l} I_1 = I - I_2 \\ R_3E_2 - R_1E_1 = I_1R_2(R_1+R_3) + R_3R_1(\underbrace{I - I_2}_{I_1}) \end{array} \right.$$

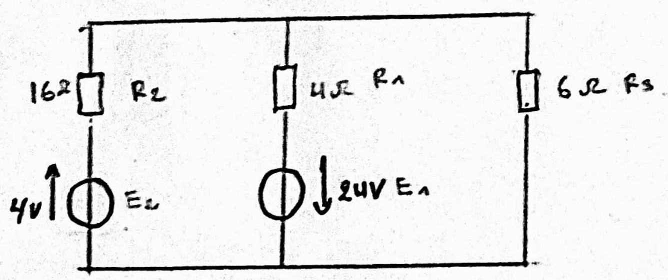
$$\begin{cases} I_1 = I - I_2 \\ R_3 E_2 - R_1 E_2 = I_1 R_2 (R_1 + R_3) + R_3 R_1 I_1 \end{cases}$$

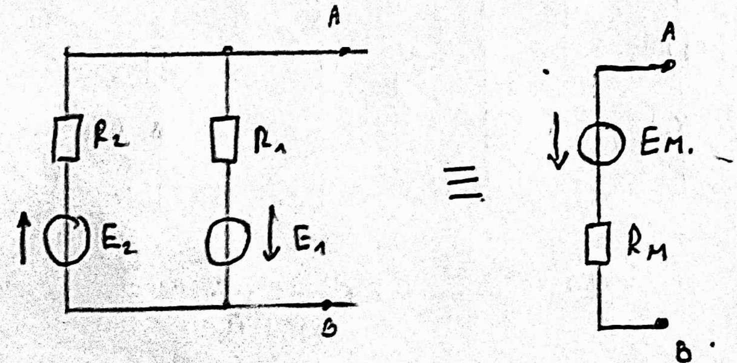
2) $$\boxed{I_1 = \frac{E_2 R_3 - E_1 R_1}{R_2 (R_1 + R_3) + R_3 R_1}}$$

- théorème de Millman



$$\equiv$$





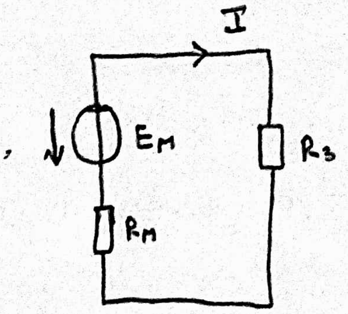
$$\frac{1}{R_M} = \sum \frac{1}{R_i} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_M} = \frac{R_1 + R_2}{R_1 R_2}$$

$$\boxed{R_M = \frac{R_1 R_2}{R_1 + R_2}}$$

$$E_M = \frac{\sum \frac{E_i}{R_i}}{\sum \frac{1}{R_i}} = \frac{\frac{E_1}{R_1} - \frac{E_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$

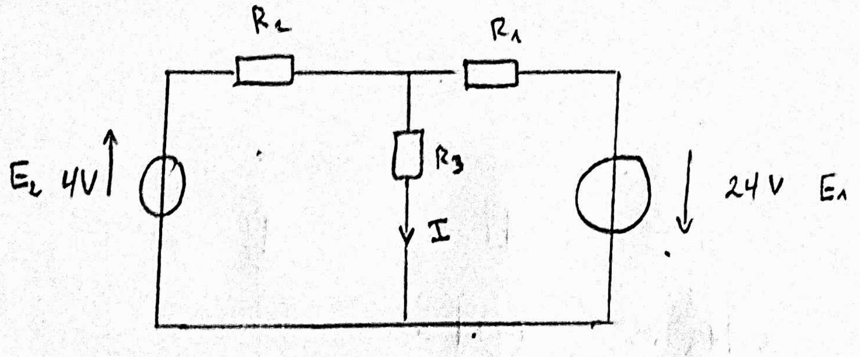
$$E_M = \frac{R_2 E_1 - R_1 E_2}{R_1 R_2 \times \left(\frac{R_1 + R_2}{R_1 \times R_2}\right)} = \boxed{\frac{E_1 R_2 - E_2 R_1}{R_1 + R_2}}$$



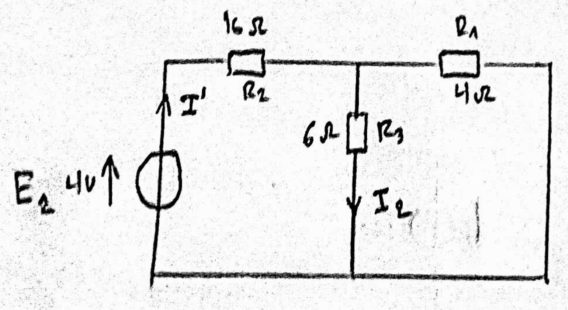
$$I = \frac{E_M}{R_M + R_3} = \frac{E_1R_2 - E_2R_1}{(R_1+R_2)\times\left(\frac{R_1R_2}{R_1+R_2} + R_3\right)}$$

$$\boxed{I = \frac{E_1R_2 - E_2R_1}{R_1R_2 + R_1R_3 + R_2R_3}}$$

- théorème de superposition.



$$\equiv \quad I = I_2 - I_1.$$



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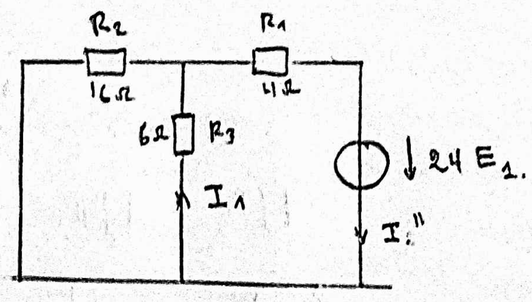


$$I' = \frac{E_2}{R_{eq}} \quad , \quad R_{eq} = \frac{R_1R_3}{R_1+R_3} + R_2.$$

$$I'' = \frac{E}{R_{eq}} \quad , \quad R_{eq} = \frac{R_2R_3}{R_2+R_3} + R_1.$$

$$I_1 = \frac{R_2}{R_2+R_3} \times I''$$

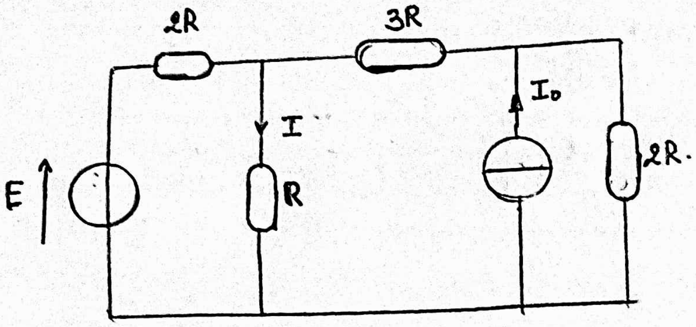
$$\boxed{I_2 = \frac{R_1}{R_1+R_3} \times \frac{E_2}{\left(\frac{R_1R_3}{R_1+R_3} + R_2\right)}}$$

$$\boxed{I_1 = \frac{R_2}{R_2+R_3} \times \frac{E_1}{\left(\frac{R_2R_3}{R_2+R_3} + R_1\right)}}$$

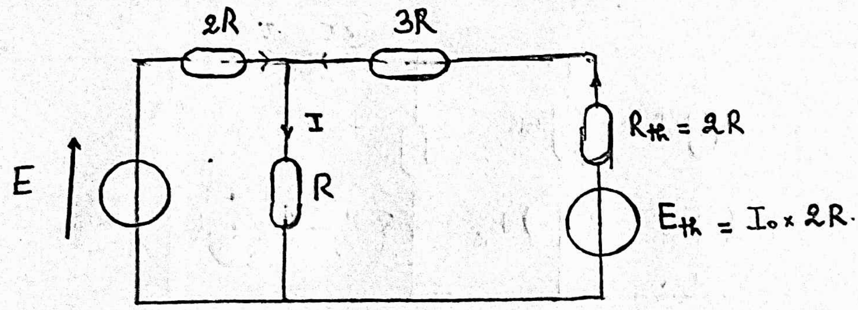
$$I_2 = \frac{R_1 E_2}{R_1R_3 + R_1R_2 + R_3R_2} \qquad I_1 = \frac{R_2 E_1}{R_2R_3 + R_1R_2 + R_1R_3}$$

$$I = I_2 - I_1 \Rightarrow \boxed{I = \frac{E_2 R_1 - E_1 R_2}{R_1R_3 + R_1R_2 + R_2R_3}}$$

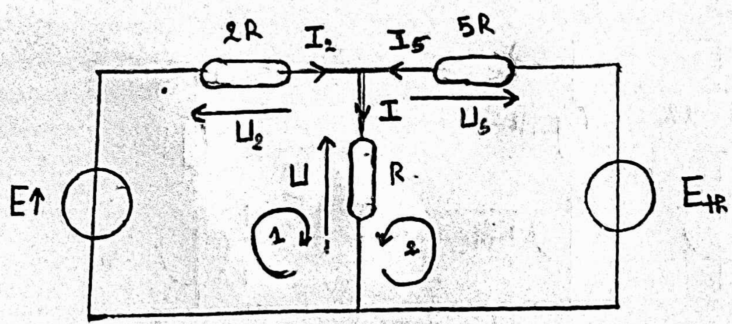
Exercice 3



1)



2- a-



$$\begin{cases} I = I_2 + I_5 \\ E - U_2 - U = 0 \\ E_{th} - U_5 - U = 0 \end{cases}$$

$$\begin{matrix} L_1 \\ L_2 \\ L_3 \end{matrix}\begin{cases} I = I_2 + I_5 \\ E - 2RI_2 - RI = 0 \\ E_{th} - 5RI_5 - RI = 0 \end{cases} \Rightarrow \begin{matrix} \\ 5L_2 \rightarrow \\ 2L_3 \rightarrow \end{matrix}\begin{cases} I = I_2 + I_5 \\ 5E - 10RI_2 - 5RI = 0 \\ 2E_{th} - 10RI_5 - 2RI = 0 \end{cases}$$

$$L_1 \begin{cases} I = I_2 + I_5 \\ 5E + 2E_{th} - 10R(I_2+I_5) - 7RI = 0 \end{cases} \; L_2 + L_3$$

$$\Rightarrow \begin{cases} I = I_2 + I_5 \\ 5E + 2E_{th} - 10RI - 7RI = 0 \end{cases} \Rightarrow \begin{cases} I = I_2 + I_5 \\ 5E + 2E_{th} - 17RI = 0 \end{cases}$$

$$I = \frac{5E + 2E_{th}}{17R}$$

$$I = \frac{5E + 2(I_0 \times 2R)}{17R}$$

$$\boxed{I = \frac{5E + 4RI_0}{17R}}$$

b).

* théorème de thévenin

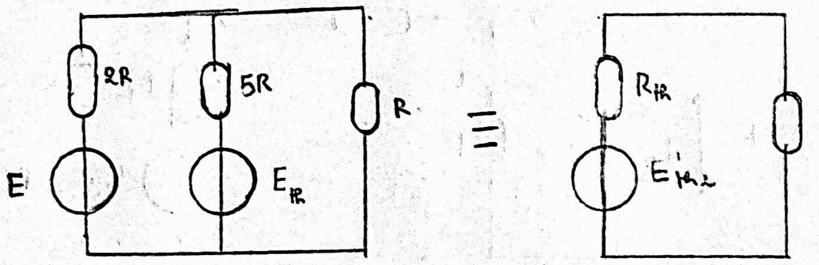
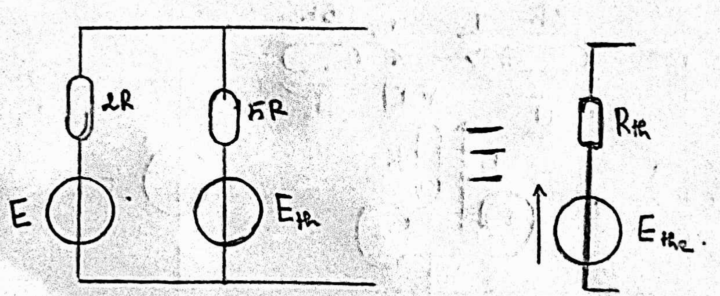


Figure ①

avec



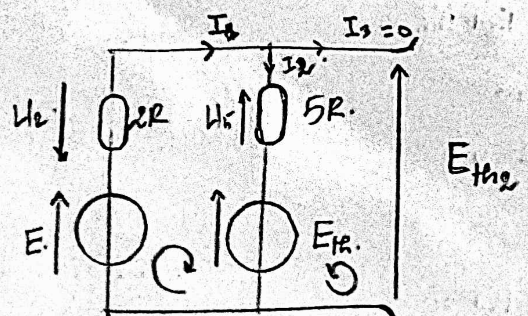


Figure ②

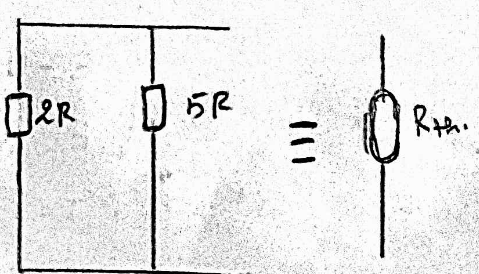


Figure ③

$E_{th_2} = ??$ Figure ②

$$\begin{cases} I_1 = I_2 + I_3 \\ E - U_2 - U_5 - E_{th} = 0 \\ E_{th_2} - U_5 - E_{th} = 0 \end{cases} \Rightarrow \begin{cases} I_1 = I_2 \\ E - 2RI_1 - 5RI_1 - E_{th} = 0 \\ E_{th_2} - 5RI_1 - E_{th} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} I_1 = I_2 \\ E - E_{th} = 7RI_2 \\ E_{th_2} = 5RI_1 + E_{th} \end{cases} \Rightarrow \begin{cases} I_1 = I_2 \\ I_1 = \dfrac{E - E_{th}}{7R} \\ E_{th_2} = 5R\left(\dfrac{E - E_{th}}{7R}\right) + E_{th} \end{cases}$$

$$E_{th_2} = \frac{5E - 5E_{th}}{7} + E_{th} = \boxed{E_{th_2} = \frac{5E + 2E_{th}}{7}}$$

$R_{th} = ??$ Figure ③

$$\frac{1}{R_{th}} = \frac{1}{2R} + \frac{1}{5R}$$

$$\frac{1}{R_{th}} = \frac{5+2}{10R} \Rightarrow \boxed{R_{th} = \frac{10R}{7}}$$

$I = ??$ Figure ①

$$R_{eq} = R_{th} + R = \frac{10R}{7} + R$$

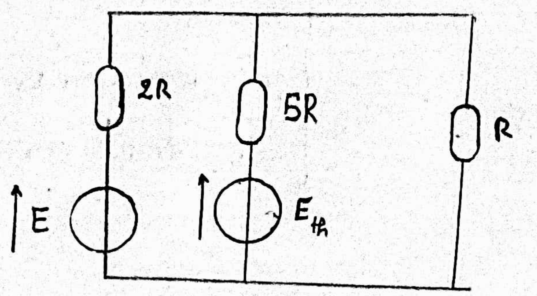
$$\boxed{R_{eq} = \frac{17R}{7}}$$

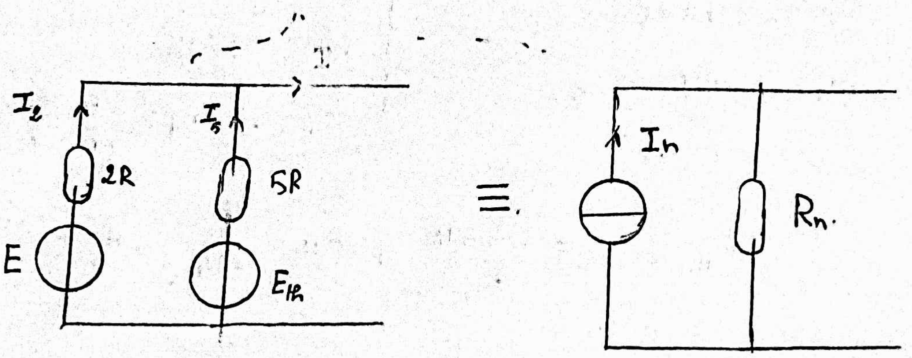
$$I = \frac{E_{th_2}}{R_{eq}} = \frac{\dfrac{5E + 2E_{th}}{7}}{\dfrac{17R}{7}}$$

$$I = \frac{5E + 2E_{th}}{17R} =$$

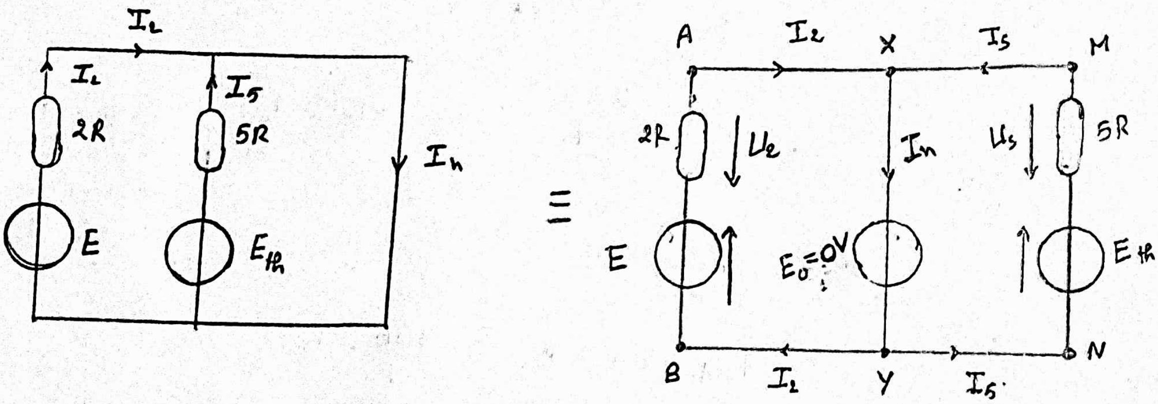
$$\boxed{I = \frac{5E + 4I_0R}{17R}}$$

⊛ théorème de Norton.





$\underline{\underline{I_n = ??}}$



$U_{AB} = U_{xy} = U_{MN}$, $U_{xy} = E_0 = 0V$

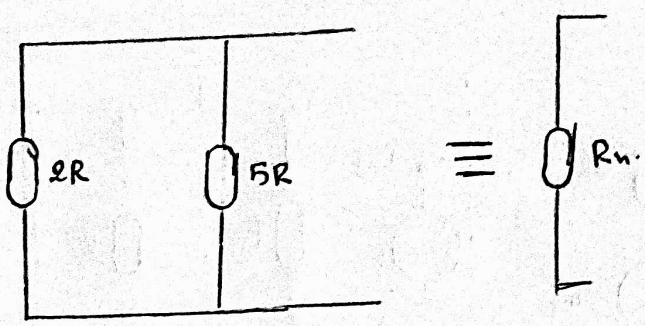
$$\begin{cases} U_{AB} = 0 \\ U_{MN} = 0 \\ U_{xy} = 0V \end{cases} \Rightarrow \begin{cases} E - U_2 = 0 \\ E_{th} - U_5 = 0 \\ E_0 = 0V \end{cases}$$

$$\Rightarrow \begin{cases} E = U_2 \\ E_{th} = U_5 \\ E_0 = 0V \end{cases} \Rightarrow \begin{cases} E = 2R \times I_2 \\ E_{th} = 5R \times I_2 \\ E_0 = 0V \end{cases} \Rightarrow \begin{cases} I_2 = \frac{E}{2R} \\ I_5 = \frac{E_{th}}{5R} \\ E_0 = 0V \end{cases}$$

$$I_n = I_2 + I_5 = \frac{E}{2R} + \frac{E_{th}}{5R}$$

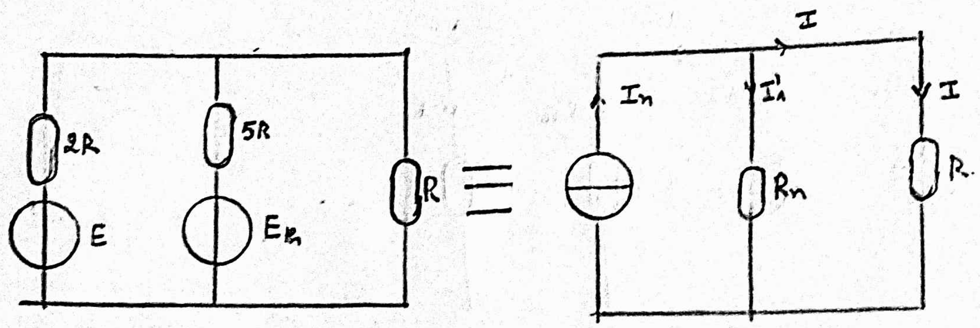
$$\boxed{I_n = \frac{5E + 2E_{th}}{10R}}$$

$\underline{R_n = ??}$



$$\frac{1}{R_n} = \frac{1}{2R} + \frac{1}{5R} = \frac{7}{10R}$$

$$\boxed{R_n = \frac{10R}{7}}$$



diviseur de courant : $I = \frac{R_n}{R_n + R} I_n$

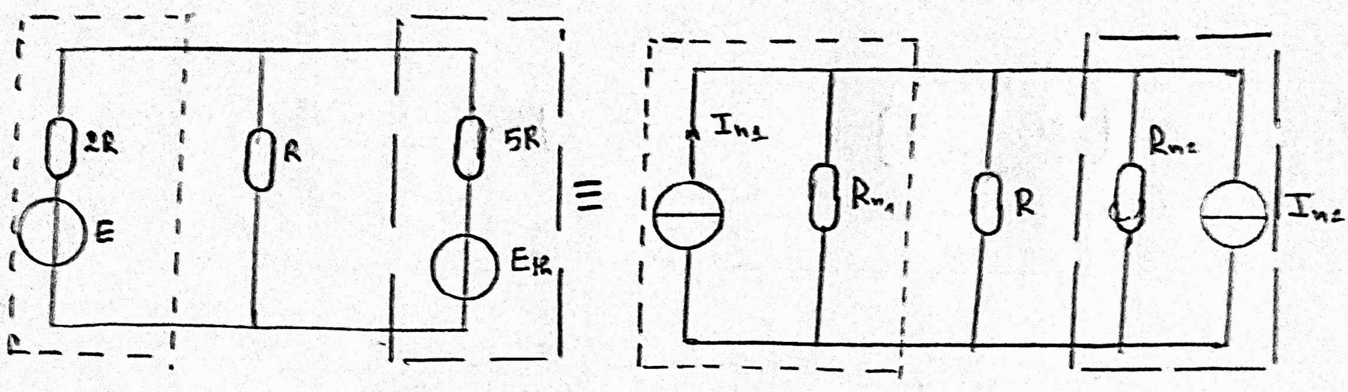
$$I = \frac{\frac{10\cancel{R}}{7}}{\frac{10R}{7} + R} \times \frac{(5E + 2E_{th})}{10\cancel{R}}$$

$$= \frac{1}{\cancel{7} \times \frac{10R}{\cancel{7}} + 7R} \times \frac{5E + 2E_{th}}{1}$$

$$= \frac{5E + 2E_{th}}{17R}$$

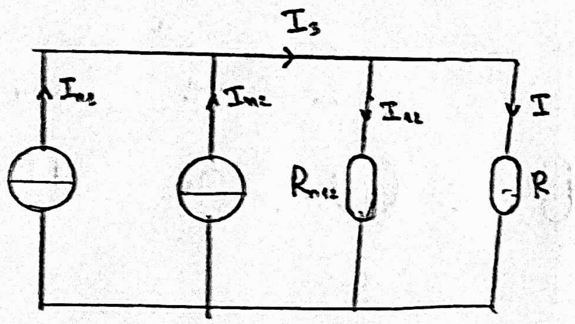
$$I = \frac{5E + 4I_0R}{17R}$$

c) La transformation Thévenin - Norton.



$$I_{n1} = \frac{E}{2R} \qquad I_{n2} = \frac{E_{th}}{5R}$$

$$R_{n1} = 2R \qquad // \qquad R_{n2} = 5R$$



$$I_3 = I_{n1} + I_{n2} \Rightarrow I_3 = \frac{E}{2R} + \frac{E_{th}}{5R} \Rightarrow \boxed{I_3 = \frac{5E + 2E_{th}}{10R}}$$

$R_{n1} // R_{n2} \Rightarrow \frac{1}{R_{n12}} = \frac{1}{R_{n1}} + \frac{1}{R_{n2}} = \frac{R_{n1} + R_{n2}}{R_{n1} R_{n2}}$

$$R_{n12} = \frac{R_{n1} R_{n2}}{R_{n1} + R_{n2}} \qquad R_{n12} = \frac{2R \times 5R}{2R + 5R} \qquad R_{n12} = \frac{10R^2}{7R} \qquad \boxed{R_{n12} = \frac{10R}{7}}$$

diviseur de courant

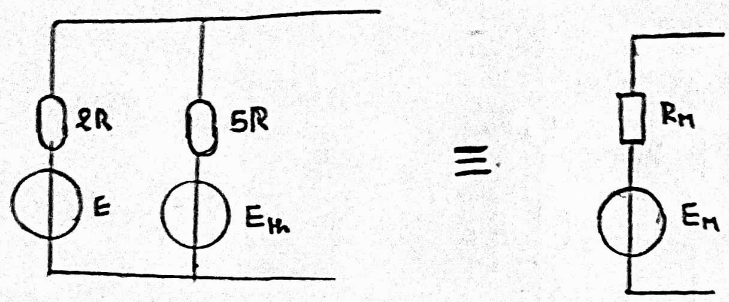
$$I = \frac{R_{n12}}{R_{n12} + R} I_3$$

$$I = \frac{\frac{10R}{7}}{\frac{10R}{7} + R} \times \frac{5E + 2E_{th}}{10R}$$

$$I = \frac{5E + 2E_{th}}{17R}$$

$$\boxed{I = \frac{5E + 4I_0R}{17R}}$$

d- théorème de Millman



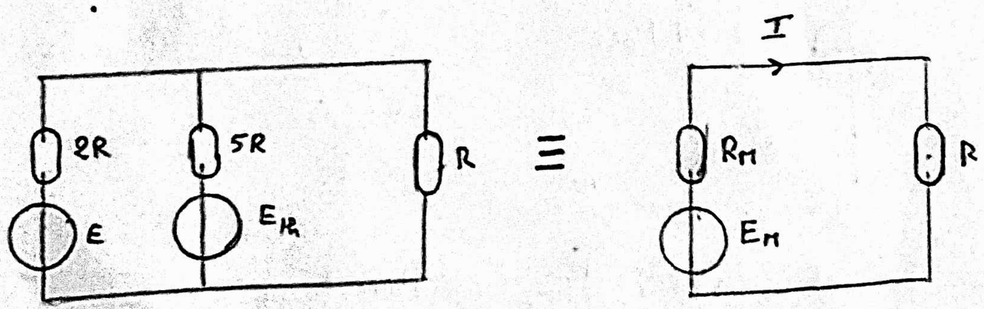
$$E_M = \frac{\sum \frac{E_i}{R_i}}{\sum \frac{1}{R_i}} \qquad \cdot \ \frac{1}{R_M} = \sum \frac{1}{R_i}$$

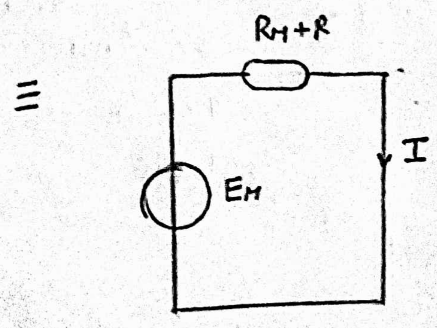
$$= \frac{\frac{E_{th}}{5R} + \frac{E}{2R}}{\frac{1}{5R} + \frac{1}{2R}} \qquad , \ \frac{1}{R_M} = \frac{1}{5R} + \frac{1}{2R} = \frac{2+5}{10R} = \frac{7}{10R}$$

$$\Rightarrow \boxed{R_M = \frac{10R}{7}}$$

$$\frac{\frac{2E_{th} + 5E}{10R}}{\frac{7}{10R}}$$

$$\boxed{E_M = \frac{2E_{th} + 5E}{7}}$$



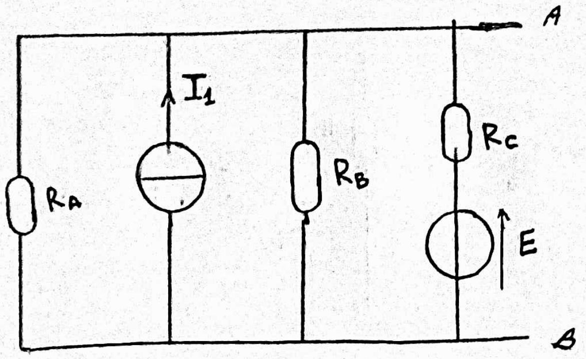


$$I = \frac{E_M}{R_M + R}$$

$$I = \frac{\frac{5E + 2E_{th}}{7}}{\frac{10R}{7} + R} \qquad I = \frac{5E + 2E_{th}}{17R}$$

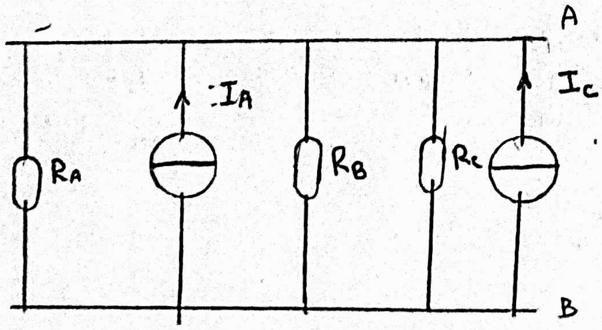
$$\boxed{I = \frac{5E + 4I_0R}{17R}}$$

Exercice 4.



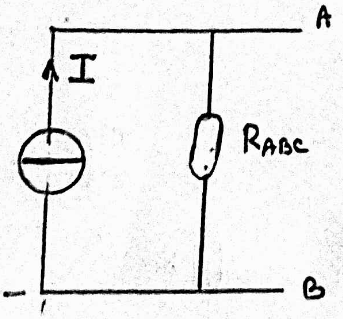
Modèle de Norton.

$\equiv$



$R_A // R_B // R_c$, $I_A // I_c$

$\equiv$



$$I = I_A + I_c = I_A + \frac{E}{R_c}$$

$$\frac{1}{R_{ABC}} = \frac{1}{R_A} + \frac{1}{R_B} + \frac{1}{R_C}$$

$$I = 3\times10^{-3} + \frac{12}{3\times10^{3}}$$
$$= 3\times10^{-3} + 4\times10^{-3}$$
$$= 7\times10^{-3}\,A.$$
$$\boxed{I = 7mA}$$

$$R_{ABC} = \frac{1}{\frac{1}{R_A} + \frac{1}{R_B} + \frac{1}{R_C}} = \frac{1}{\frac{1}{1,5} + \frac{1}{1} + \frac{1}{3}}$$
$$= \frac{1}{\frac{2}{3} + \frac{1}{3} + \frac{3}{3}} = \frac{3}{6} = \frac{1}{2}\,k\Omega$$
$$\boxed{R_{ABC} = 500\Omega}$$

Modèle de thévenin

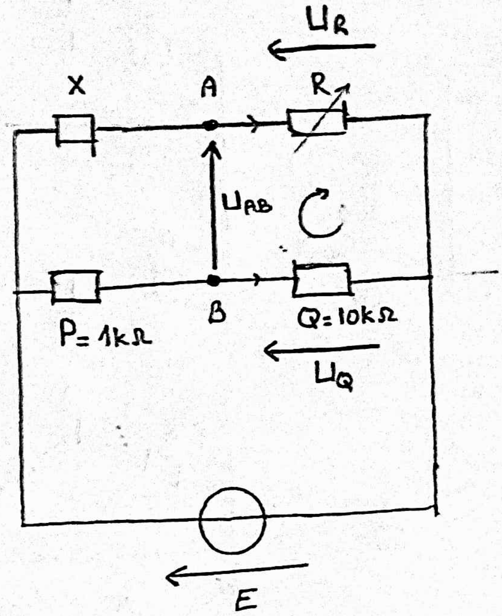
$\equiv$

$\boxed{R_{ABC} = 500\Omega}$

$E_{th} = R_{ABC} \times I = 500 \times 7\times10^{-3} = 3,5V$

$\boxed{E_{th} = 3,5V}$

Exercice 5



Modèle de thévenin

- $E_{th} = U_{AB}$

$U_{AB} - U_R + U_Q = 0$

$\boxed{U_{AB} = U_R - U_Q}$

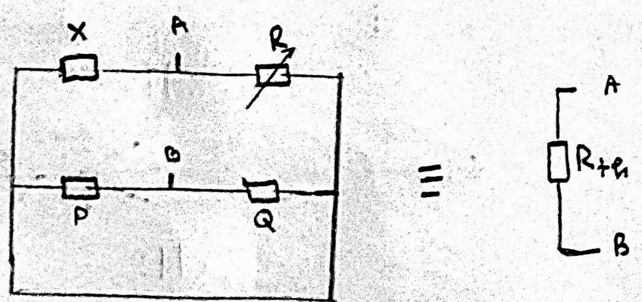
diviseur de tension.

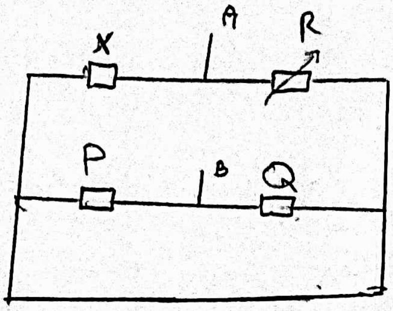
$U_R = \frac{R}{R+X} E$, $U_Q = \frac{Q}{Q+P} E$

$E_{th} = U_{AB} = U_R - U_Q = \frac{R}{R+X} E - \frac{Q}{Q+P} E$

$E_{th} = \left(\frac{R}{R+X} - \frac{Q}{Q+P} \right) E$

- $R_{th} = ??$





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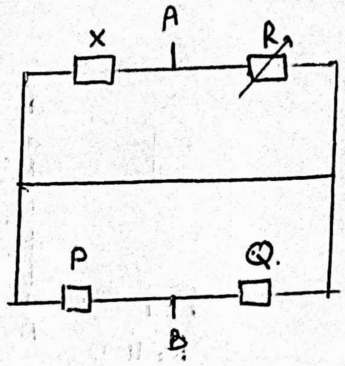

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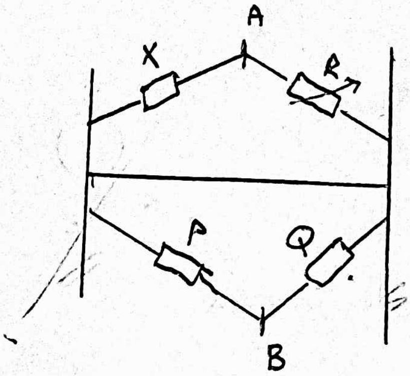

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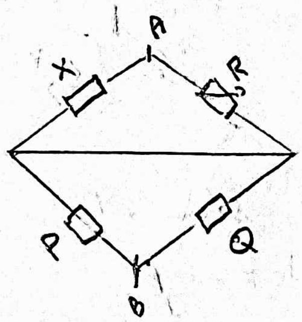
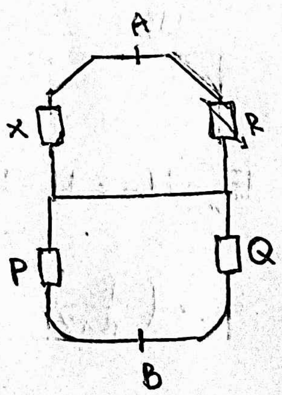


$$\frac{1}{R_{e1}} = \frac{1}{X} + \frac{1}{R}$$

$$\boxed{R_{e1} = \frac{XR}{X+R}}$$

$$\frac{1}{R_{e2}} = \frac{1}{P} + \frac{1}{Q}, \quad \boxed{R_{e2} = \frac{PQ}{P+Q}}$$

$\equiv$



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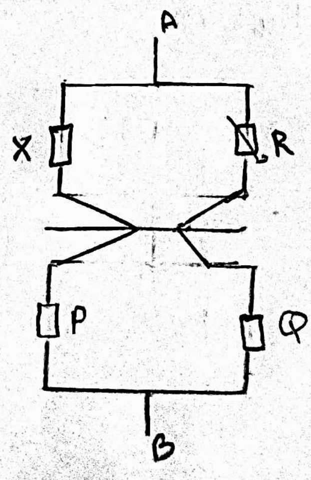
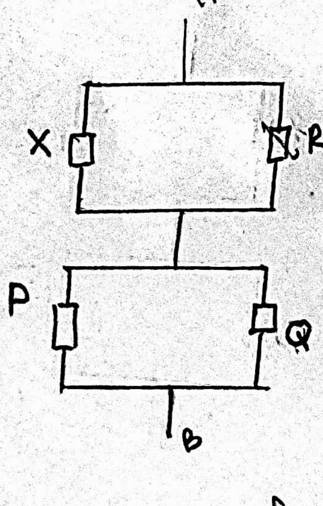
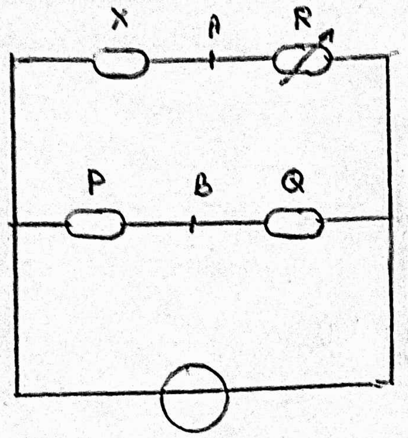




$\equiv$

$$R_{th} = R_{e1} + R_{e2}$$

$$\boxed{R_{th} = \frac{XR}{X+R} + \frac{PQ}{P+Q}}$$



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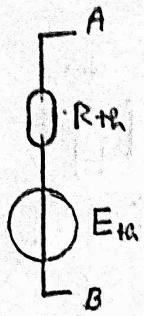


$$\boxed{\begin{aligned} E_{th} &= U_{AB} = \left(\frac{R}{R+X} - \frac{Q}{Q+P}\right)E \\ R_{th} &= \frac{XR}{X+R} + \frac{PQ}{P+Q} \end{aligned}}$$

$\underline{\underline{U_{AB} = 0V\ ??}}$

$$U_{AB} = 0V \Rightarrow \left(\frac{R}{R+X} - \frac{Q}{Q+P}\right)E = 0$$

$$\Rightarrow \frac{R}{R+X} - \frac{Q}{Q+P} = 0$$

$$\Rightarrow \frac{R}{R+X} = \frac{Q}{Q+P}$$

$$R = R\left(\frac{Q}{Q+P}\right) + X\left(\frac{Q}{Q+P}\right)$$

$$R\left(1 - \frac{Q}{Q+P}\right) = X\left(\frac{Q}{Q+P}\right)$$

$$R = \frac{\frac{XQ}{\cancel{Q+P}}}{\frac{\cancel{Q}+P-\cancel{Q}}{\cancel{Q+P}}}$$

$$\boxed{U_{AB} = 0V \Leftrightarrow R = \frac{XQ}{P}}$$

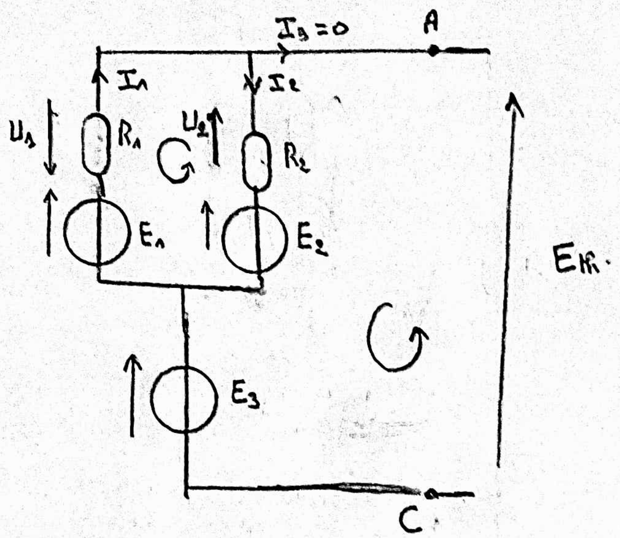
A.N $\quad X = \frac{PR}{Q} = \frac{1 \times 8{,}75}{10}$

$X = 0{,}875\ k\Omega$

$$\boxed{X = 875\,\Omega}$$

Exercice 6

1).



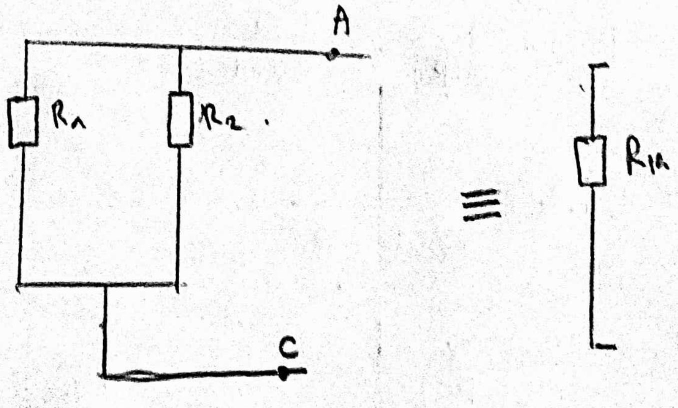
$$\begin{cases} E_2 + U_2 + U_1 - E_1 = 0 \\ E_{th} + U_2 - E_2 - E_3 = 0 \\ I_1 = I_2 \end{cases} \Rightarrow \begin{cases} E_2 + R_2 I_2 + R_1 I_1 - E_1 = 0 \\ E_{th} - R_2 I_2 - E_2 - E_3 = 0 \\ I_1 = I_2 \end{cases}$$

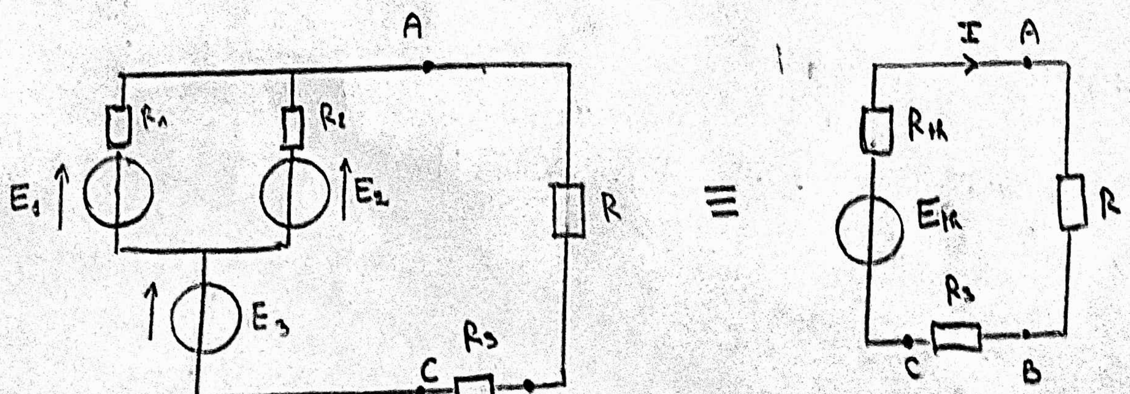
$$\Rightarrow \begin{cases} E_2 + I_2(R_2 + R_1) - E_1 = 0 \\ E_{th} = E_3 + E_2 + R_2 I_2 \end{cases} \Rightarrow \begin{cases} I_2 = \dfrac{E_1 - E_2}{R_1 + R_2} \\ E_{th} = E_3 + E_2 + R_2\left(\dfrac{E_1 - E_2}{R_1 + R_2}\right) \end{cases}$$

$$\boxed{E_{th} = E_3 + E_2 + R_2\left(\frac{E_1 - E_2}{R_1 + R_2}\right)}$$

$$\frac{1}{R_{th}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\boxed{R_{th} = \frac{R_1 R_2}{R_1 + R_2}}$$





$$I = \frac{E_{th}}{R_{th} + R + R_3}$$

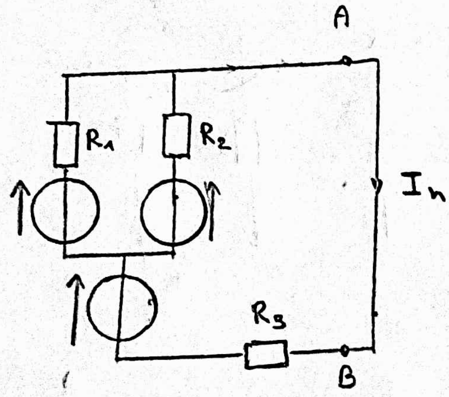
$$= \frac{E_3 + E_2 + R_2\left(\frac{E_1 - E_2}{R_1 + R_2}\right)}{\frac{R_1 R_2}{R_1 + R_2} + R + R_3}$$

$$= \frac{2 + 1 + 2\left(\frac{3-1}{2+2}\right)}{\frac{2\times 2}{2+2} + 5 + 2}$$

$$= \frac{3 + 2\left(\frac{2}{4}\right)}{1 + 5 + 2} = \frac{4}{8} = \frac{1}{2}\,A$$

$$\boxed{I = \frac{1}{2}\,A}$$

2) théorème de Norton



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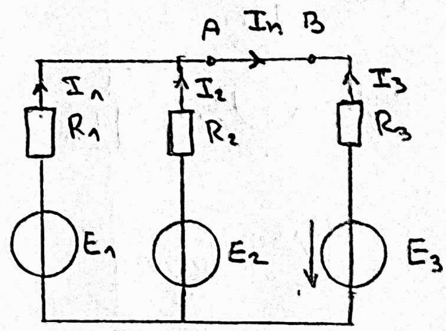
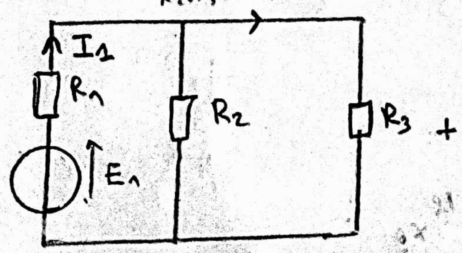


$I_1 = \frac{E_1}{R_1 + \frac{R_2 R_3}{R_2 + R_3}}$ I_4

$I_2 = \frac{E_2}{R_2 + \frac{R_1 R_3}{R_1 + R_3}}$ I_5

I_6



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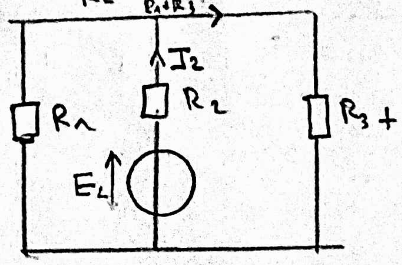
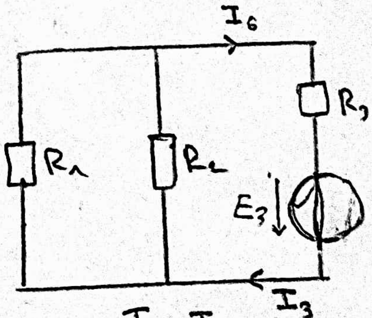


+



$$I_4 = \frac{R_2}{R_2 + R_3} \times I_1$$

$$= \frac{R_2}{R_2 + R_3} \times \frac{E_1}{R_1 + \frac{R_2 R_3}{R_2 + R_3}}$$

$$I_5 = \frac{R_1}{R_1 + R_3} \times I_2$$

$$= \frac{R_1}{R_1 + R_2} \times \frac{E_2}{R_2 + \frac{R_1 R_3}{R_1 + R_3}}$$

$$I_6 = I_3$$

$$=$$

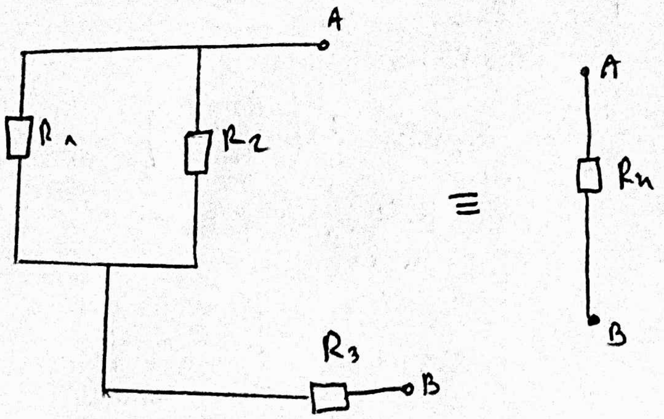
$$I_n = \frac{1}{2} \times \frac{3}{2+1} + \frac{1}{2} \times \frac{1}{2+1} + \frac{2}{1+2}$$

$$= \frac{1}{2} + \frac{1}{6} + \frac{2}{3}$$

$$\boxed{I_n = \frac{4}{3}\,A}$$

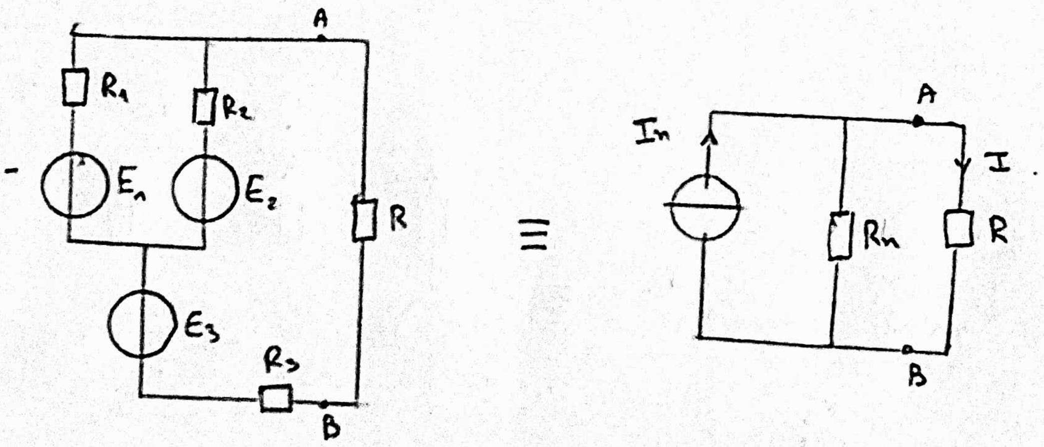
$$\boxed{I_n = \frac{R_2}{R_2 + R_3} \times \frac{E_1}{R_1 + \frac{R_2 R_3}{R_2 + R_3}} + \frac{R_1}{R_1 + R_2} \times \frac{E_2}{R_2 + \frac{R_1 R_3}{R_1 + R_3}} + \frac{E_3}{\frac{R_1 R_2}{R_1 + R_2} + R_3}}$$

$$R_{e1} = \frac{R_1 R_2}{R_1 + R_2}$$



$$R_n = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

$$\boxed{R_n = 3\Omega}$$



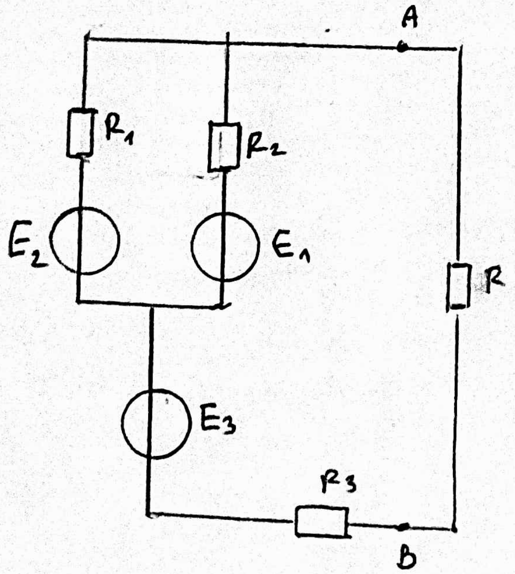
$$I = \frac{R_n}{R_n + R} I_n$$

$$I = \frac{3}{3+5} \times \frac{4}{3} = \frac{4}{8} A = \frac{1}{2} A.$$

$$\boxed{I = \frac{1}{2} A}$$

3) transformation Thévenin-Norton

=>



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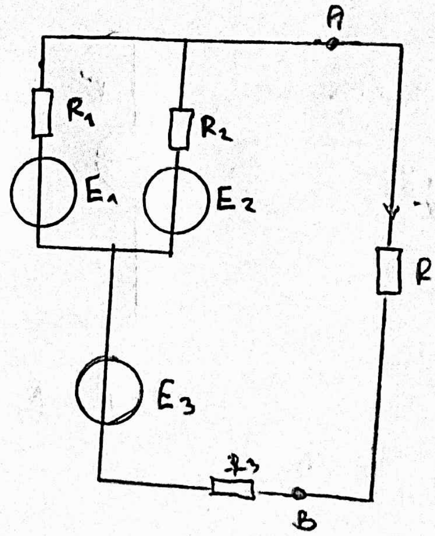
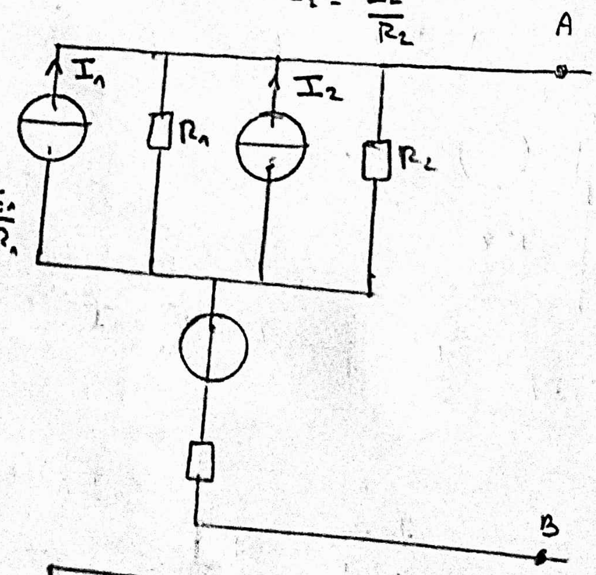


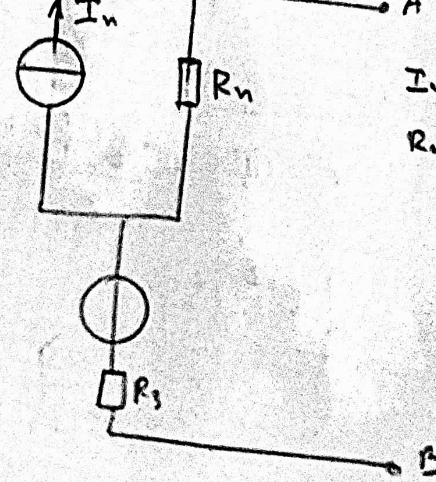
≡

$I_2 = \frac{E_2}{R_2}$



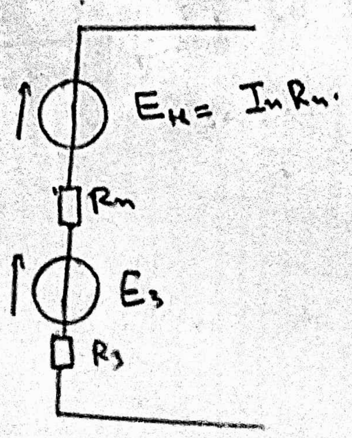
$I_1 = \frac{E_1}{R_1}$

≡

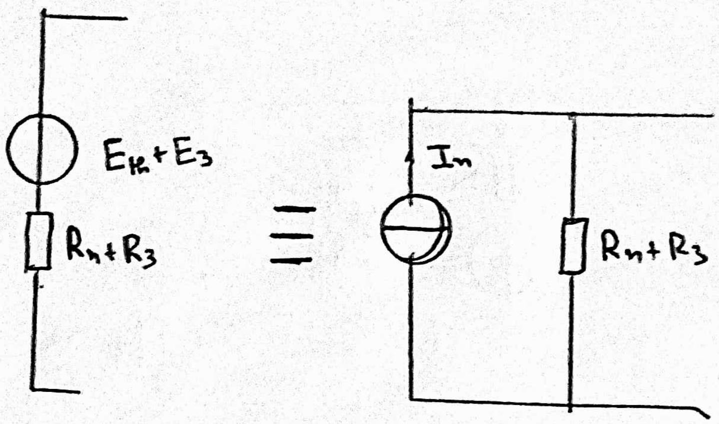


$I_n = I_1 + I_2.$

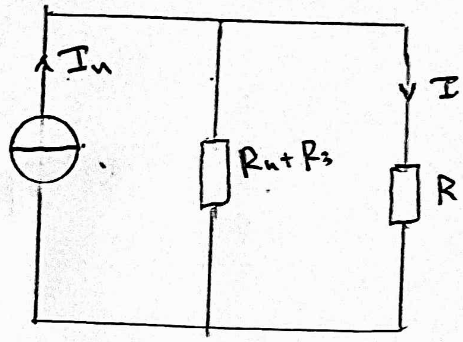
$R_n = \frac{R_1 R_2}{R_1 + R_2}.$



$E_N = I_n R_n.$



$$I_n = \frac{E_{th}+E_3}{R_n+R_3}$$



$$I = \frac{R_n+R_3}{R+R_n+R_3} \times \frac{E_{th}+E_3}{R_n+R_3}$$

$$= \frac{\frac{R_1R_2}{R_1+R_2}+R_3}{R+\frac{R_1R_2}{R_1+R_2}+R_3} \times \frac{I_nR_n+E_3}{\frac{R_1R_2}{R_1+R_2}+R_3}$$

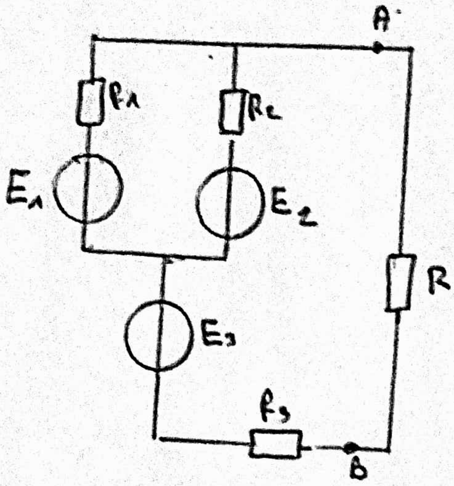
$$= \frac{1+2}{5+1+2} \times \frac{(I_1+I_2)\times 1+2}{1+2}$$

$$= \frac{\cancel{1+2}}{8} \times \frac{\frac{E_1}{R_1}+\frac{E_2}{R_2}+2}{\cancel{3}}$$

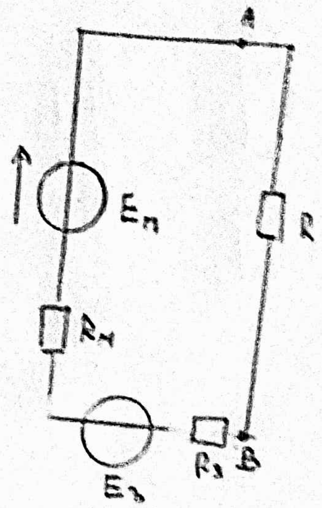
$$= \frac{\frac{3}{2}+\frac{1}{2}+2}{8} = \frac{4}{8}A$$

$$\boxed{I = \frac{1}{2}A}$$

Théorème de Millman



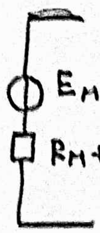
$\equiv$



$$E_M = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{E_1R_2 + E_2R_1}{R_1R_2 \times \left(\frac{R_1R_2}{R_1+R_2}\right)} = \frac{3\times 2 + 2}{4} = \frac{8}{4} = 2V$$

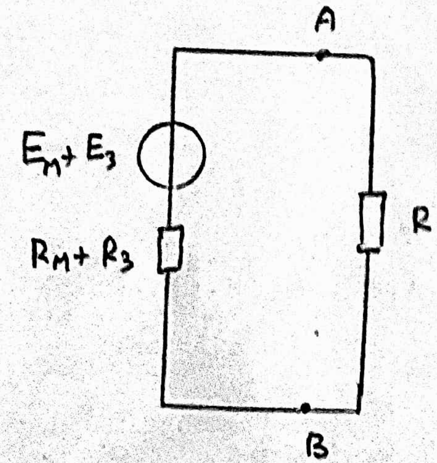
$$\frac{1}{R_M} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \boxed{R_M = \frac{R_1R_2}{R_1+R_2}} = 1\Omega$$

$\equiv$



$E_M + E_3 = 2 + 2 = 4V$

$R_M + R_3 = 1 + 2 = 3\Omega$



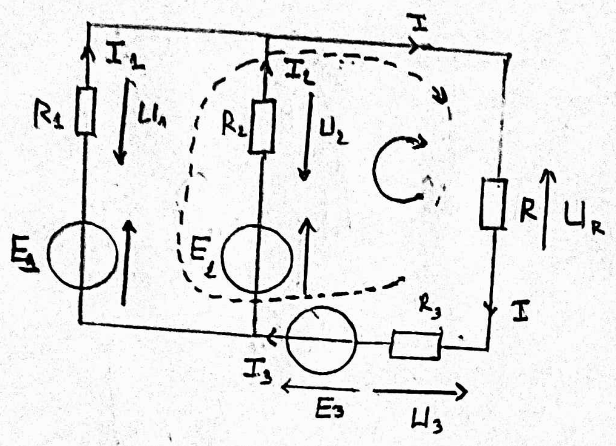
$\Rightarrow$

$$I = \frac{E_M + E_3}{R_M + R_3}$$

$$I = \frac{4}{3+1} = \frac{1}{2} A.$$

$$\boxed{I = \frac{1}{1} A}$$

Lois de Kirchhoff :



$$\begin{cases} I_1 + I_2 = I \\ -U_3 + E_3 + E_2 - U_2 - U_R = 0 \\ -U_3 + E_3 + E_1 - U_1 - U_R = 0 \end{cases}$$

$$\begin{array}{l} L_1 \\ L_2 \\ L_3 \end{array} \begin{cases} I = I_1 + I_2 \\ E_3 + E_2 - R_3 I - R_2 I_2 - RI = 0 \\ E_3 + E_1 - R_3 I - R_1 I_1 - RI = 0 \end{cases}$$

$$\begin{array}{l} L_1 \\ R_1 L_2 \to \\ R_2 L_3 \to \end{array} \begin{cases} I = I_1 + I_2 \\ R_1 E_3 + R_1 E_2 - R_1 R_3 I - R_1 R_2 I_2 - R_1 R I = 0 \\ R_2 E_3 + R_2 E_1 - R_2 R_3 I - R_2 R_1 I_1 - R_2 R I = 0 \end{cases}$$

$$L_1 + L_2 = R_1(E_3 + E_2) + R_2(E_3 + E_1) - I(R_1 R_3 + R_2 R_3) - R_1 R_2 (\underbrace{I_1 + I_2}_{I}) - I(RR_1 + RR_2) = 0$$

$$= R_1(E_3 + E_2) + R_2(E_3 + E_1) - I(R_1 R_3 + R_2 R_3) - I R_1 R_2 - I(RR_1 + RR_2) = 0$$

$\Rightarrow$

$$I = \frac{R_1(E_3 + E_2) + R_2(E_3 + E_1)}{R_1 R_3 + R_2 R_3 + R_1 R_2 + RR_1 + RR_2}$$

$$= \frac{2(2+1) + 2(2+3)}{4 + 4 + 4 + 5 \times 2 + 5 \times 2} = \frac{6 + 10}{32} = \frac{16}{32} = \frac{1}{2} A$$

$$\boxed{I = \frac{1}{2} A}$$